Applications of Engineering and Scientific Principles for a Gamified Household Food Waste Simulation

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Introduction

Food waste is a major global sustainability challenge that contributes to environmental degradation, economic loss, and food insecurity. Studies estimate that nearly one-third of all food produced globally is lost or wasted, with a substantial portion occurring at the household level. Household food waste often results from over-purchasing, poor meal planning, improper storage, and food spoilage. Because consumer behavior plays a central role in this issue, interactive educational tools can help individuals better understand how daily decisions influence food waste generation.

The objective of our capstone project is to develop a gamified simulation that models household food purchasing, storage, consumption, and waste behaviors. In this system, players make decisions about grocery shopping, meal preparation, and food storage, while the simulation dynamically tracks how these choices influence the amount of food consumed or discarded. By combining gameplay elements with scientifically grounded models, the prototype aims to increase user awareness of sustainable food management practices.

Engineering and scientific principles are essential for ensuring that the simulation produces realistic and educational outcomes. Mathematical models allow the system to simulate food inventory changes, spoilage processes, and waste generation in a way that reflects real-world household dynamics. This report identifies key engineering principles applied to the prototype, demonstrates how these principles can be implemented through quantitative models and calculations, and proposes design innovations that enhance the effectiveness of the simulation.

Component/Process Selection

The component selected for engineering analysis is the household food inventory and waste generation algorithm. This algorithm functions as the core system that governs how food flows through the simulated household environment.

Within the prototype, the food inventory system performs several essential functions. First, it tracks food items entering the household through grocery purchases. Each item is assigned attributes such as quantity, storage time, and spoilage rate. Second, the system updates the status of each food item daily by evaluating whether the item is consumed, remains in storage, or

becomes waste due to spoilage. Finally, the simulation records waste generation and calculates performance metrics that provide feedback to the player.

This process can be represented as a system flow model (Figure 1), where food enters the system as purchased items, moves into the household inventory, and eventually exits the system through consumption or disposal. Treating the household as an engineering system with defined inputs, outputs, and internal states enables the use of quantitative modeling techniques to ensure realistic behavior within the simulation.

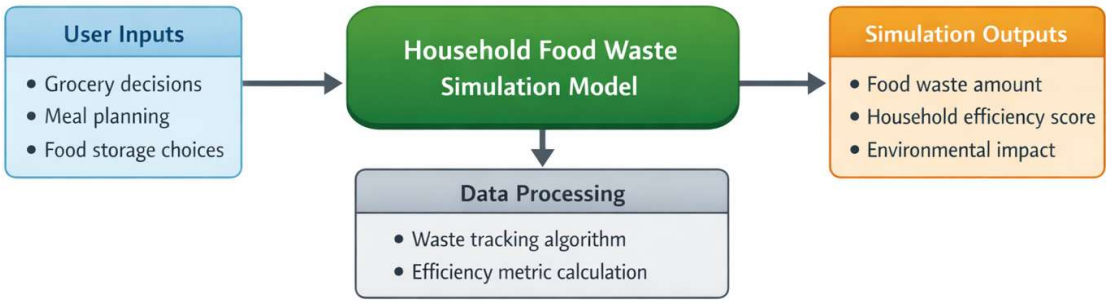

Figure 1. Household food waste simulation system architecture.

This diagram illustrates the overall structure of the simulation, showing how user inputs such as grocery purchasing decisions, meal planning, and food storage behaviors interact with the simulation model. The system processes these inputs using algorithms that track food inventory, simulate spoilage, and calculate waste generation. The outputs of the model include the amount of food waste produced and a household efficiency score that provides feedback on the player's performance.

The simulation models household food management as a sequence of decisions made by the user, including meal planning, grocery purchasing, food storage, and meal preparation. These decisions determine whether food is consumed efficiently or becomes waste. The decision-making process implemented in the simulation is illustrated in Figure 2.

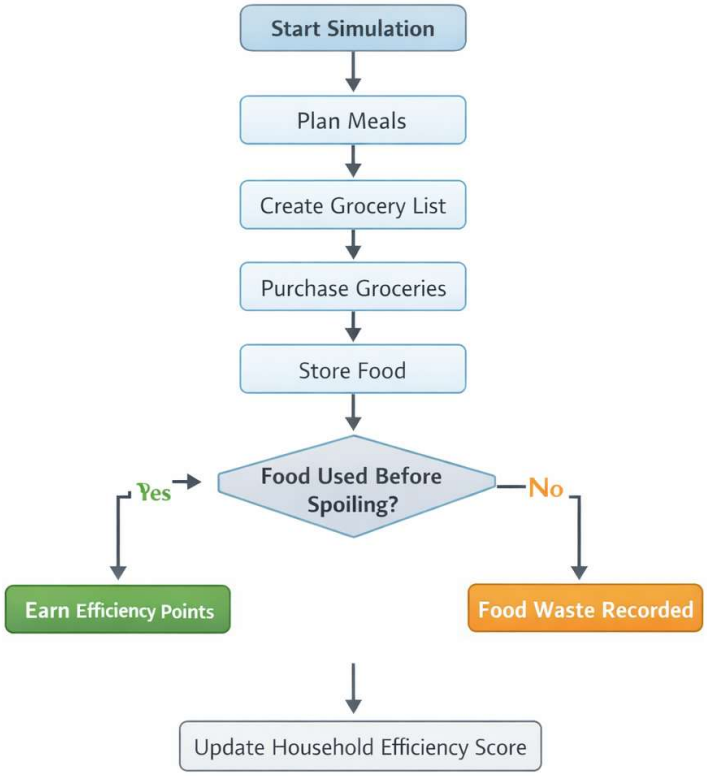

Figure 2. Decision-making flow within the household food waste simulation.

This flowchart represents the sequence of actions a player performs during the simulation. Users plan meals, create grocery lists, purchase food, store items, and prepare meals. At each stage, the system evaluates whether food is consumed before spoilage occurs. Successful decisions increase the player's efficiency score, while unused or spoiled food is recorded as waste within the simulation model.

Engineering and Scientific Principles

Several engineering and scientific principles provide the theoretical foundation for modeling food waste generation in the simulation.

Conservation of Mass (Mass Balance)

One of the most fundamental principles applied to the simulation is the conservation of mass, which states that matter cannot be created or destroyed within a closed system. In engineering applications, mass balance equations are used to track material flows through systems such as chemical reactors, water treatment plants, and environmental processes.

In the context of household food management, the mass balance equation ensures that all food entering the system is accounted for through consumption, waste, or storage accumulation. The governing equation is:

$$F_p - F_c - F_w = \frac{dFs}{dt}$$

Where:

Fp = food purchased (kg/week)
Fc = food consumed (kg/week)
Fw = food wasted (kg/week)
Fs = food stored in the household inventory

The conservation of mass principle is critical to the simulation because it ensures that the food inventory system behaves in a physically realistic way. In real households, food does not disappear or appear spontaneously; every item that enters the home must eventually be consumed, remain in storage, or become waste. By implementing the mass balance equation, the simulation can accurately track the flow of food through the system and maintain consistency between purchases, consumption, and waste generation.

Without this principle, the simulation could produce unrealistic results, such as food being consumed without being purchased or inventory levels becoming negative. The mass balance framework also allows the system to quantify how household decisions affect food waste accumulation over time. For example, if a player consistently purchases more food than they consume, the model will show an increasing accumulation of stored food that eventually leads to spoilage and waste. In this way, the conservation of mass provides the structural foundation for the simulation's food tracking system, ensuring that the model remains both scientifically valid and logically consistent.

Food Spoilage Kinetics

Food spoilage is modeled using first-order exponential decay, a common approach in food science and chemical engineering for describing degradation processes.

The spoilage model is expressed as:

$$N(t) = N_0 e^{-kt}$$

Where:

N0 = initial food quality
N(t) = food quality at time ttt

k = spoilage rate constant
t = time in days

Food quality decreases exponentially over time. Once the quality drops below a critical threshold value Ncrit, the food item is considered spoiled and is categorized as waste in the simulation.

Modeling food spoilage using first-order decay kinetics is important because it provides a scientifically grounded way to simulate the deterioration of food quality over time. In real-world food systems, perishable items gradually degrade due to microbial growth, enzymatic reactions, and chemical changes. These processes often follow exponential decay patterns, making first-order kinetics a reasonable approximation for modeling spoilage behavior.

In the context of the simulation, the spoilage model determines when food items transition from usable products to waste. This allows the game to realistically simulate the consequences of delayed consumption or poor storage practices. For instance, if a player purchases large quantities of perishable food but does not consume them quickly enough, the exponential decay model will reduce the quality value of those items until they fall below the waste threshold.

This principle is also important because it introduces time-dependent dynamics into the system. Food items in the inventory do not remain static; their condition continuously changes as the simulation progresses. This encourages players to prioritize consuming foods with shorter shelf lives and reinforces the educational objective of promoting better food management practices.

Probabilistic Consumption Model

Household food consumption patterns are inherently variable and influenced by factors such as daily schedules, appetite, and meal planning. To represent this uncertainty, the simulation incorporates probabilistic modeling.

The probability that a food item is consumed on a given day can be approximated as:

$$P_c = \frac{C_d}{F_s}$$

Where:

Pc = probability of consumption
Cd = daily household consumption rate
Fs = total food inventory

Using this probability, the simulation employs a random number generator to determine whether individual food items are consumed during each simulation cycle. This stochastic approach introduces behavioral variability that better reflects real-world household dynamics.

Probabilistic modeling is essential because household food consumption patterns are inherently unpredictable. In reality, food consumption varies depending on daily schedules, meal planning, appetite, and other behavioral factors. A deterministic model in which food is consumed at fixed intervals would fail to capture this variability and would produce unrealistic outcomes.

By incorporating probability into the simulation, the system can introduce randomness that better reflects real-world behavior. Each day, the simulation evaluates the probability that certain foods will be consumed, allowing consumption patterns to vary over time. This stochastic approach makes the simulation more dynamic and prevents players from predicting outcomes with absolute certainty.

The use of probabilistic modeling also improves the educational value of the game. Players must make strategic decisions about how much food to purchase and when to consume it, knowing that real-life variability may influence outcomes. This reflects real household decision-making and highlights how small behavioral differences can significantly impact food waste generation.

Dynamic Systems Modeling

The household food management process can be modeled as a **dynamic system**, where system variables evolve over time based on user decisions and internal processes.

Inputs to the system include:

- grocery purchases
- meal planning decisions
- storage choices

Outputs include:

- food consumption
- food waste generation

Dynamic systems modeling is important because it allows the simulation to track long-term trends rather than isolated events. For example, repeated over-purchasing will gradually increase the amount of stored food, which eventually leads to higher spoilage rates.

This system's approach allows players to observe how small behavioral decisions accumulate over time and influence the overall efficiency of household food management.

This feature would allow players to observe how proper storage can significantly extend shelf life and reduce waste.

Treating the household food system as a dynamic system is important because it allows the simulation to represent interactions between multiple variables over time. Household food management involves a continuous cycle of purchasing, storing, consuming, and discarding food. These processes influence one another and evolve as time progresses.

Dynamic systems modeling allows the simulation to track how changes in one variable affect the entire system. For example, an increase in food purchases will expand the household inventory, which may lead to higher consumption or increased spoilage depending on player behavior. By modeling these interactions, the simulation can demonstrate long-term patterns in food waste generation rather than focusing only on individual decisions. Figure 2 provides a visualization of how the decisions affect the generation.

This system's perspective is particularly valuable for educational purposes. Players can observe how their purchasing habits and consumption strategies affect food waste levels across multiple simulated weeks or months. As a result, the simulation provides insight into how small behavioral adjustments can produce significant long-term reductions in household food waste.

Practical Application and Quantitative Model

To illustrate how these principles are implemented in the prototype, consider a simulated household that purchases 6 kilograms of food per week.

Assume the following weekly outcomes:

Food purchased: 6 kg
Food consumed: 4.5 kg
Food spoiled: 1.2 kg

Applying the mass balance equation:

$$6 - 4.5 - 1.2 = 0.3$$

This calculation indicates that 0.3 kilograms of food remain in the household inventory, which carries over into the next simulation cycle.

Spoilage Model Calculation

Consider a food item with an initial quality level:

$$N0 = 100$$

Assume the spoilage constant:

$$k = 0.20$$

After five days:

$$N(t) = 100e^{-0.20(5)}$$

$$N(t) = 100e^{-1}$$

$$N(t) = 36.8$$

If the spoilage threshold is defined as:

$$Ncrit = 40$$

then the food item is classified as spoiled and converted into food waste within the simulation.

If the spoilage threshold is defined as Ncrit=40, then the food item is classified as spoiled once its quality falls below this threshold. As shown in Figure 3, food quality decreases exponentially over time. When the quality value falls below the critical threshold, the item is categorized as food waste within the simulation. Table 1 provides the example data used to obtain Figure 3.

Table 1. Example data used in exponential decline in food quality over time predicted by the first-order decay model.

Time (days)	Food Quality (N(t))
0	100.0
1	81.9
2	67.0
3	54.9
4	44.9
5	36.8
6	30.1
7	24.7
8	20.2
9	16.5

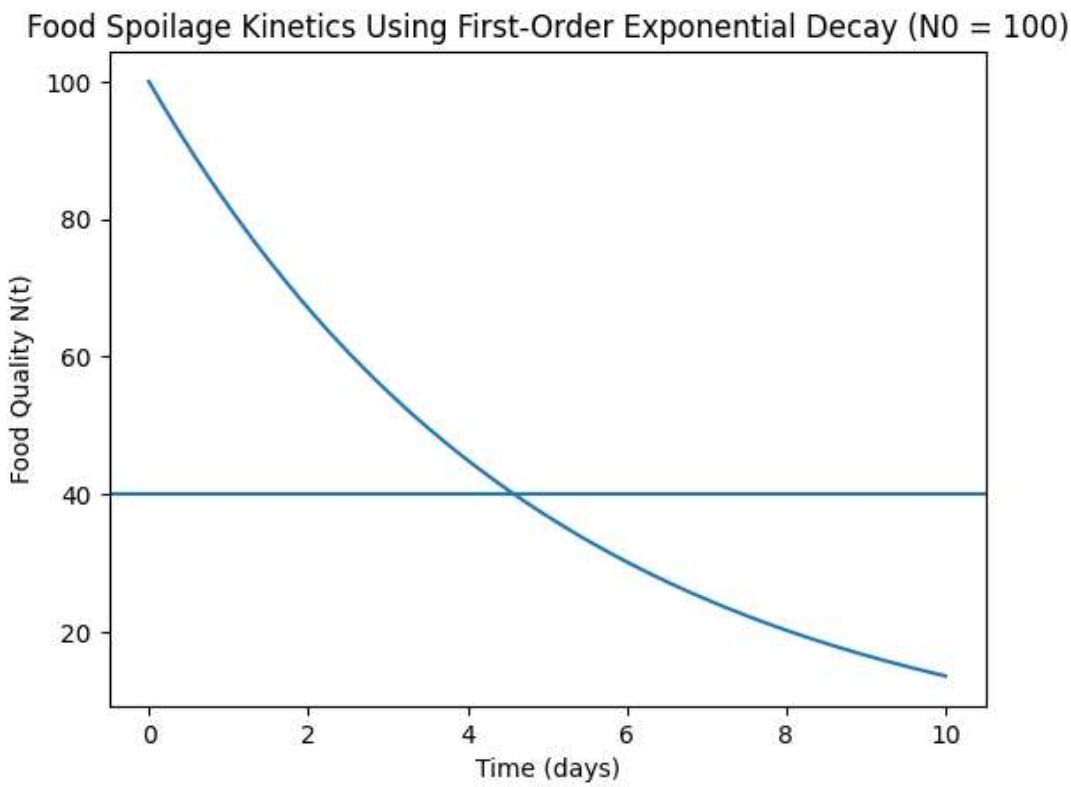

Figure 3. Food spoilage kinetics modeled using a first-order exponential decay equation.

Food quality N(t) decreases from an initial value of N0=100 with a spoilage constant k=0.20. The horizontal line represents the spoilage threshold Ncrit=40, below which the food item is classified as waste in the simulation.

Innovative Design Solutions

Applying engineering principles to the prototype led to several potential improvements that could enhance the realism and educational value of the simulation.

Dynamic Storage Conditions

One proposed innovation is to allow players to control food storage conditions within the simulation. Storage environments such as refrigeration or freezing would affect the spoilage rate constant k.

For example:

$$k_{refrigerated} = 0.10$$

$$k_{room} = 0.25$$

By modifying spoilage rates based on storage choices, the simulation can demonstrate how refrigeration significantly extends food shelf life.

Household Efficiency Metric

To reinforce sustainable food management behaviors, the simulation incorporates a household efficiency metric that is directly linked to a player scoring system. This metric evaluates how effectively a household utilizes the food it purchases by measuring the relationship between food consumption, food waste, and decision-making behaviors within the simulation.

The base efficiency metric is defined as:

$$Efficiency = \frac{F_C}{F_P}$$

Where:

Fc = total food consumed (kg)
Fp = total food purchased (kg)

This ratio represents the fraction of purchased food that is successfully consumed rather than wasted. An efficiency value close to 1 indicates highly efficient food management, while lower values indicate higher levels of waste.

For example, if a household purchases 6 kg of food and consumes 5 kg:

$$\text{Efficiency} = \frac{5}{6} = 0.83$$

This indicates that 83% of purchased food was utilized, while the remaining 17% contributed to waste or excess storage.

While this efficiency ratio provides a quantitative measure of food utilization, the simulation also incorporates behavior-based point rewards to encourage specific sustainable practices. Players earn points for making decisions that are known to reduce food waste in real households. These behaviors include planning meals before shopping, purchasing only the items on a grocery list, storing food in appropriate conditions, and successfully consuming food before it spoils.

The total score for a simulation cycle can therefore be expressed as:

$$Score = w1Pm + w2Ps + w3Pc + w4E$$

Where:

Pm = points for meal planning and grocery list adherence
Ps = points for proper food storage decisions
Pc = points for cooking and consuming available food
E = household efficiency metric
w1, w2, w3, w4 = weighting factors that determine the importance of each behavior

For example, if a player plans meals, stores food properly, and consumes most of their purchases, the combined point system will reward them with a higher score. This structure allows the simulation to quantify both behavioral decisions and overall system efficiency, linking engineering metrics with game mechanics. This is modeled by Figure 4.

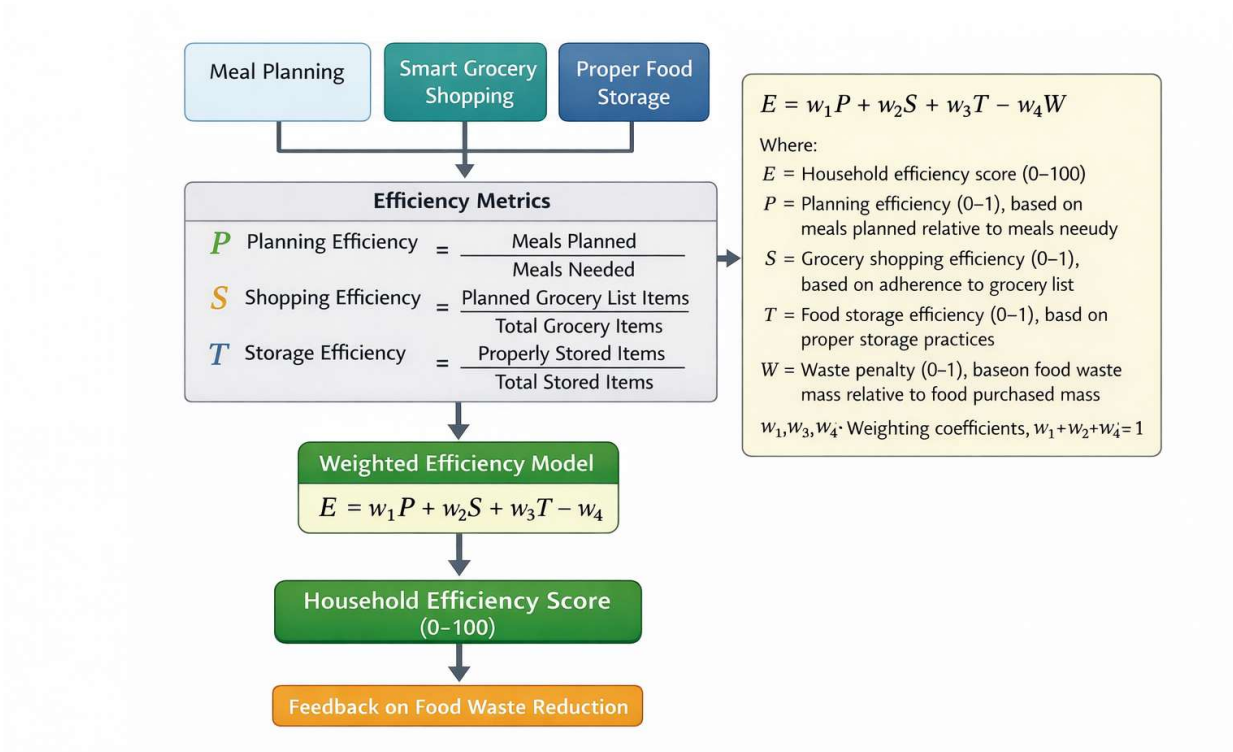

Figure 4. Household efficiency scoring model used in the simulation.

This diagram illustrates how player decisions are translated into quantitative performance metrics. Actions such as meal planning, grocery shopping behavior, and food storage practices are converted into efficiency variables that are incorporated into a weighted scoring equation. The resulting household efficiency score reflects how effectively the simulated household minimizes food waste and provides feedback to guide improved decision-making.

Integrating this scoring model into the prototype serves two important purposes. First, it provides a measurable performance indicator that reflects how effectively players manage household food resources. Second, it reinforces positive behaviors by rewarding actions that reduce food waste. By combining quantitative efficiency calculations with gamified incentives, the simulation

encourages players to adopt more sustainable food management strategies while maintaining an engaging and interactive gameplay experience.

Collaborative Reflection

Integrating engineering principles into the prototype provided valuable insights into how theoretical concepts can guide practical system design. By applying mass balance equations, spoilage kinetics, and probabilistic modeling, the team was able to construct a simulation that realistically represents household food management dynamics.

One challenge encountered during development was balancing model accuracy with computational simplicity. Real food spoilage processes depend on numerous environmental variables, but incorporating all of these factors would complicate the simulation and reduce gameplay accessibility. To address this issue, the team decided to adopt simplified models that capture the essential behavior of food degradation while remaining efficient for real-time simulation.

Another challenge involved translating engineering models into intuitive gameplay mechanics. The gamified scoring system was developed as a solution to bridge this gap, allowing quantitative efficiency metrics to influence player rewards and decision-making.

Future improvements to the prototype will include calibrating model parameters using real household food waste datasets. This will allow the simulation to produce more accurate predictions and enhance its effectiveness as an educational tool for reducing food waste.

In addition to the engineering principles used to model household food systems, the design of the simulation also considers how users interact with and adopt educational technologies. One framework that helps explain user adoption is the Technology Acceptance Model III (TAM III). This model proposes that two primary factors influence whether individuals are willing to use a technological system: perceived usefulness and perceived ease of use.

Perceived usefulness refers to the extent to which a user believes that a system helps them accomplish a task or improve performance. In the context of the household food waste simulation, usefulness is achieved by providing clear feedback on the outcomes of a player's decisions. For example, the simulation calculates a household efficiency metric and awards points for sustainable behaviors such as meal planning, purchasing only necessary groceries, properly storing food, and consuming food before it spoils. By showing players how these decisions reduce food waste, the simulation demonstrates the practical value of sustainable food management practices.

Perceived ease of use refers to how simple and intuitive a system is for users to understand and interact with. The gamified structure of the simulation supports this principle by breaking complex sustainability concepts into manageable tasks such as grocery shopping, storage decisions, and cooking meals. Players receive immediate feedback through the scoring system and waste metrics, allowing them to quickly understand the consequences of their actions.

Incorporating these concepts into the design of the simulation helps ensure that the system is not only scientifically grounded but also engaging and accessible to users. By combining quantitative modeling with user-centered design principles, the simulation aims to encourage sustained interaction and promote long-term awareness of household food waste behaviors.